BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Debugging and Nested Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does sequence mean in programming?

2. What is 4 + 3 * 2?

A) 10 B) 14
C) 20 D) 24

3. What is 2 ^ 3 in OCR ERL?

4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

Today's objectives

Identify and fix syntax and logic errors. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Syntax error

An incorrect language keyword or missing ENDIF breaks the grammar.

Logic error

A running program may produce an incorrect result because its steps are wrong.

Nested selection

Each IF within an IF needs the correct closing ENDIF.

Debugging cycle

Test, locate, correct and retest the fault.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Syntax error	An incorrect language keyword or missing ENDIF breaks the grammar. For example, consider a membership checker.
Logic error	A running program may produce an incorrect result because its steps are wrong. For example, consider a sports bands.
Nested selection	Each IF within an IF needs the correct closing ENDIF. For example, consider a ticket discounts.
Debugging cycle	Test, locate, correct and retest the fault. For example, consider a club rules.

I Do - One worked example

A membership checker is being developed. Explain how syntax error and logic error are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

3. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

4. Give one accurate point and a linked reason for each technique.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A sports bands is being programmed. Explain why syntax error matters. [2 marks]

2. For a ticket discounts, explain how logic error is used. [2 marks]

3. A club rules needs nested selection and debugging cycle. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the logical operator so 260 is rejected.

```
octet = 260
if octet >= 0 OR octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how logic error and nested selection can help a ride height. [4 marks]

Q4. Explain the roles of syntax error, nested selection and debugging cycle when creating a reward scoring. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An incorrect language keyword or missing ENDIF breaks the grammar. ☐ T ☐ F
2. An incorrect language keyword or missing ENDIF breaks the grammar ☐ T ☐ F
3. A running program may produce an incorrect result because its steps are wrong ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Instructions execute in the specified order. ☐

Q2 A - 10 ☐

Q3 8; ^ denotes exponentiation in OCR ERL. ☐

Q4 Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly. ☐

Code First - Model

```
octet = 260
if octet >= 0 AND octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

Output: Invalid

We Do - Check the model

1. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1]. ☐

2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1]. ☐

3. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Syntax error [1]; Nested selection [1]. ☐

Q3 Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. ☐

Q4 Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: